



UOW
COLLEGE
AUSTRALIA

—
PATHWAYS TO
UNIVERSITY OF
WOLLONGONG

Subject Outline

DPEN022: Enabling Mathematics B

Autumn 2026

Wollongong

Credit Points: 8

Subject Pre-requisite: HSC General Mathematics OR Year 10 Advanced Mathematics

Subject Co-requisite: Nil

Equivalence: MATH140

Exclusions: MATH151

University Entrance Program

Foundation Studies (CRICOS Course Code: 007732G)

Foundation Studies Extended Program (CRICOS Course Code: 023266F)

Diploma of Engineering (CRICOS Course Code: 0101925, 0101924, 083351A)

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Subject Information

Subject Coordinators

Refer to subject Moodle page

Subject Description

DPEN022 continues the development of essential mathematical knowledge begun in DPEN021, supporting students progressing into the Faculty of Engineering and Information Sciences and the Bachelor of Secondary Education (Mathematics). The emphasis of this subject is on extending mathematical competence and strengthening problem-solving skills. The main areas of study include trigonometry, limits, differentiation, and integration. Engineering applications are incorporated where suitable to demonstrate how these mathematical tools are used in real-world contexts.

DPEN021 provides the foundation and is a prerequisite for DPEN022.

Subject Learning Outcomes

On successful completion of this subject, students will be able to:

1. Demonstrate knowledge of important mathematical concepts such as, algebra, trigonometry, limits and calculus.
2. Select and apply appropriate mathematical skills and techniques in solving problems.
3. Use and interpret appropriate terminology, diagrams, and symbols in mathematical contexts.

Subject Delivery Mode

This subject is delivered as an “on campus” subject, however scheduled classes, learning activities and assessments may include a combination of face-to-face and/or online learning. Students must be prepared and available to attend the nominated UOW College campus whenever required. Where you live or travel from, conflicting work commitments or recreational activities are not considered valid reasons for not attending learning or assessment activities on campus. See more on attendance requirements below.

Students with an approved Access Plan or Academic Consideration that includes an alternative arrangement to attending on campus classes or assessments, must confirm these arrangements with the Subject Coordinator.

UOW College may need to modify teaching locations, teaching delivery, and assessment delivery at short notice in response to unforeseen circumstances such as health or environmental factors. Students will be advised in a timely manner via SOLS if this occurs.

Attendance and Workload

Attending scheduled classes (face-to-face or online) is an important aspect of your learning, helping you to understand subject content and develop skills. For this reason, attendance at lectures and tutorials at UOW College is compulsory and you must attend a minimum of 80% of all classes. Students who do not meet this requirement may be awarded a Technical Fail (TF) grade for the subject, regardless of the total final mark. Attendance is also an important factor in determining whether a student is a genuine student or a student deemed to be at risk of failing to meet course progress requirements. For details, refer to the [UOW College Attendance Policy](#) and the [Course Progression Policy](#).

The credit point value of a subject reflects the workload required for students to successfully complete the subject. Each credit point represents 1.5 to 2 hours of work including attending the scheduled classes (online or face-to-face) and self-directed study (homework, revision, research and assessment tasks) per week for a subject delivered in a standard session. For example, in a 6 credit point subject expect to spend 9 to 12 hours a week attending classes and completing related study.

Key References

There is no set textbook for this subject. Students will be able to access class notes and class exercises to accompany the weekly topics on Moodle.

To access material online, go to the subject Moodle site accessible via SOLS.

Recommended resources:

New Senior Mathematics Advanced for Years 11 and 12, Third Edition by J.B. Fitzpatrick and B.Aus. ISBN: 978 1 4886 1829 1

Recommended readings are not intended as an exhaustive list and students should use the [Library](#) catalogue and databases to locate additional resources.

Calculators

Students will be required to bring a calculator to all classes.

Only calculators on the UOW College list of approved calculators will be permitted in exams. The list of approved calculators is located at [Calculators](#) on the UOW website. Approved calculators must have a tamperproof label. The list of where the label can be obtained and other calculator information is also on the website.

Subject Structure & Timetable

DPEN022 – Enabling Mathematics B is a 4-credit point subject delivered as 8 hours of scheduled classes per week. The subject is delivered through 4 x 2-hour classes over a 6-week session.

The 6-week session is followed by a 3-week break. Deferred and Supplementary assessments will be conducted in the second week of this break (Week 8).

You can access your personal subject timetable by logging into SOLS and selecting 'My Timetable'.

Schedule of Learning Activities

Week	Learning Content	Assessment Due
Week 1	Trigonometry	
	Exact Trigonometric Ratios Angle of any Magnitude Solving Trigonometric Equations in Degrees Radian Measure Solving Trigonometric Equations in Radians	
Week 2	Trigonometry	Moodle Assignment Trigonometry Exam
	Graphs of Trigonometric Functions Pythagoras Theorem and Trigonometric Identities Limits	
Week 3	Differentiation	
	Differentiation by First Principles Differentiation of polynomials Product, Quotient and Chain Rule Equation of Tangent Line Derivative of Exponential Functions	
Week 4	Differentiation & Integration	Moodle Assignment Differentiation Exam
	Derivative of Logarithmic Functions Derivative of Trigonometric Functions The Primitive Function The Indefinite Integral The Definite Integral	
Week 5	Integration	
	Under the Curve Integration of Exponential Functions Integration of Logarithmic Functions Integration of Trigonometric Functions Mixed Integration	
Week 6	Integration	Moodle Assignment Integration Exam
	Integration by Substitution	

Assessment

Assessment Summary

No.	Assessment Name	Weight (% of final mark)	GenAI usage	Subject Learning Outcomes Assessed	Week No/due date
1	Trigonometry Exam	20%	Prohibited	1, 2, 3	Week 2C
2	Moodle Assignments	10%	Somewhat Prohibited	1, 2, 3	Week 2D Week 4D Week 6D
3	Differentiation Exam	35%	Prohibited	1, 2, 3	Week 4C
4	Integration Exam	35%	Prohibited	1, 2, 3	Week 6C

Minimum Performance Requirements

A minimum performance requirement ensures you have completed all learning and assessment to the standard required to attain the learning outcomes in this subject. Students who do not meet the minimum performance requirements, but otherwise achieve an overall mark of 50 or higher in the subject, will be awarded a Technical Fail (TF) grade for the subject.

The minimum performance requirements to pass this subject are:

- All assessment tasks must be submitted.
- *You must achieve a minimum of 40% in each of the Trigonometry Exam, Differentiation Exam and the Integration Exam.*
- A Technical Fail (TF) grade will be given where a student gains an overall total mark of 50 or greater for the subject BUT does not meet a minimum performance requirement. Students who gain an overall total mark of 50 or greater for the subject BUT do not meet a minimum performance requirement may be eligible for a supplementary exam. If the student passes the supplementary exam and meets the final exam Technical Fail requirement, the student will be given a PS 50 grade. If the student fails the supplementary exam, then the student will be given a Technical Fail (TF) grade.

Additionally, students who fail but are close to a pass may be considered for a supplementary exam. However, the overall final mark obtained for the subject and the final exam result will be considered. If eligible, a student will be given a WS grade until the results of the supplementary exams are finalised.

A final subject mark of 50% or higher is required to pass the subject. To progress through their course, or to their UOW offer, students may require a passing mark in some subjects that is higher than 50% (e.g. 60%). Students must check the Progression Guidelines in [the relevant Course Outline](#) for their course of study.

Assessment Task Details

The [Assessment and Examinations Guidelines for Students](#) outlines and formalises for students what is, and what is not, acceptable conduct related to assessments and examinations, in addition to information regarding the release of student examination results and penalties for late assessment submissions.

Assessment 1: Trigonometry Exam

Detailed information	The Trigonometry Exam will assess class notes covered in Weeks 1A – 2A. It will be a closed book, but UOW approved calculators will be allowed. You will be provided with a list of formulae, which will be available on eLearning (Moodle).
Marking Criteria	Multiple choice, short-answer and long-answer questions will be marked on demonstrated understanding of the key concepts, correctness of manipulation, clarity of argument and use of mathematical notation. A correct answer with no working may receive a mark of 0. Partial credit may be awarded for incorrect solutions. <i>You must achieve a minimum of 40% in the Trigonometry Exam. If this requirement is not met, your final mark for the subject will be a TF (Technical Fail).</i>
Length/Duration	1 hour
Weighting	20%
Due Date	Week 2C
Type of Collaboration	Individual
Gen AI usage	Prohibited: In this assessment, the use of any GenAI technology is prohibited. The content for your final submission must be your original work. Use of GenAI technology may result in an academic misconduct investigation.
Style and format	Multiple choice, short-answer and long-answer questions.
Submission method	In class
Assessment return	Results will be posted on SOLS. Students may view their Trigonometry Exam in Week 3A and/or in consultation. The exam must be returned to your teacher and cannot be kept.

Assessment 2: Moodle Assignments

Detailed information	Moodle assignments will cover material in Weeks 1A – 6A. This is an open book task, and students are encouraged to use class materials to assist them.
Marking Criteria	Each multiple choice and short answer question of the Moodle component will be marked on the correctness of the answer only. You have unlimited attempts before the closing date, and the highest mark will be recorded.
Length/Duration	2 weeks to complete. Each attempt will be open for 1 hour.
Weighting	10%
Due Date	Friday Week 2 at 11.59 pm. Friday Week 4 at 11.59 pm. Friday Week 6 at 11.59 pm.
Type of Collaboration	Individual
Gen AI usage	Somewhat Permitted: In this assessment the use of GenAI technology is permitted with acknowledgment to assist in the development of an assessment item. Permitted use includes brainstorming, planning, structuring, editing, proofreading and getting feedback on an assessment item. It is important to: • use only the permitted GenAI technology and follow instructions on how it can be used. • acknowledge the use of GenAI appropriately. • keep note of your interactions with GenAI technology and drafts of your work because your Subject Coordinator may ask you to demonstrate how you developed an assessment item. • ask your Subject Coordinator if you are unsure of assessment instructions. The content final submission for this assessment must be your original work. Misuse of GenAI technology may result in an academic misconduct investigation.
Style and format	Multiple choice questions and short answer.
Submission method	Multiple choice and short answers will be submitted into a quiz activity on Moodle.
Assessment return	Results and feedback will be available after the quiz activity has closed on Moodle.

Assessment 3: Differentiation Exam

Detailed information	The Differentiation Exam will assess class notes covered in Weeks 2D – 4A. It will be a closed book, but UOW approved calculators will be allowed. You will be provided with a list of formulae, which will be available on eLearning (Moodle).
Marking Criteria	Multiple choice, short-answer and long-answer questions will be marked on demonstrated understanding of the key concepts, correctness of manipulation, clarity of argument and use of mathematical notation. A correct answer with no working may receive a mark of 0. Partial credit may be awarded for incorrect solutions. <i>You must achieve a minimum of 40% in the Differentiation Exam. If this requirement is not met, your final mark for the subject will be a TF (Technical Fail).</i>
Length/Duration	1 hour and 20 minutes.
Weighting	35%
Due Date	Week 4C
Type of Collaboration	Individual
Gen AI usage	Prohibited: In this assessment, the use of any GenAI technology is prohibited. The content for your final submission must be your original work. Use of GenAI technology may result in an academic misconduct investigation.
Style and format	Multiple choice, short-answer and long-answer questions.
Submission method	In class
Assessment return	The Differentiation Exam is the end of session examination. End of session examinations papers are not returned to students. Students wishing to view their end of session examination paper will need to contact the subject coordinator to arrange a time for viewing.

Assessment 4: Integration Exam

Detailed information	The Integration Exam will assess class notes covered in Weeks 4D – 6A. It will be a closed book, but UOW approved calculators will be allowed. You will be provided with a list of formulae, which will be available on eLearning (Moodle).
Marking Criteria	Multiple choice, short-answer and long-answer questions will be marked on demonstrated understanding of the key concepts, correctness of manipulation, clarity of argument and use of mathematical notation. A correct answer with no working may receive a mark of 0. Partial credit may be awarded for incorrect solutions. <i>You must achieve a minimum of 40% in the Integration Exam. If this requirement is not met, your final mark for the subject will be a TF (Technical Fail).</i>
Length/Duration	1 hour and 20 minutes.
Weighting	35%
Due Date	Week 6C
Type of Collaboration	Individual
Gen AI usage	Prohibited: In this assessment, the use of any GenAI technology is prohibited. The content for your final submission must be your original work. Use of GenAI technology may result in an academic misconduct investigation.
Style and format	Multiple choice, short-answer and long-answer questions.
Submission method	In class
Assessment return	The Integration Exam is the end of session examination. End of session examinations papers are not returned to students. Students wishing to view their end of session examination paper will need to contact the subject coordinator to arrange a time for viewing.

Academic Misconduct: Plagiarism and Cheating

The UOW College expects students to submit only their own original work for assessment as a demonstration of their learning and achievement of the subject learning outcomes. The [Academic Integrity and Student Conduct Policy](#) and the [Procedure for Managing Alleged Student Misconduct](#) sets out our expectations and the process if you are alleged to have committed academic misconduct. If you plagiarise, cheat in an exam or use online tools or services to complete your work, it is likely this will be detected reported as academic misconduct.

Examples of academic misconduct include:

- Not referencing correctly (or at all) and presenting the ideas of others as your own work or without acknowledgement. Referencing is a standardised way of acknowledging sources of information and ideas used in academic work. You must learn to reference sources when undertaking assessment at the UOW College and the University.
- Re-using your own work (either in part or in full), which you have submitted previously for assessment, without appropriate acknowledgement or without the explicit permission of the Subject Coordinator.
- The use of any website or social media platform to gain access to essays or other assessment items (sometimes marketed as 'resources'). This service is sometimes at a cost to the student.
- The use of Artificial Intelligence (AI) such as 'Chat GPT' to generate answers to questions or write essays without doing so at the direction of your teacher and subject outline and without the use of these tools being acknowledged in your assessment task.
- Bringing unapproved materials into an examination either online or in person.
- Colluding with others to complete an exam or share answers either online or in person.
- Helping others to cheat either directly or indirectly, for example, by uploading an assessment item, subject outline or other course materials to a website or giving someone else your assessment item.

Academic Misconduct is routinely and frequently detected, particularly where students are having difficulties completing their own work or meeting the course workload. If you are feeling overwhelmed, need help to understand your subject content, assessment task requirements or you are having difficulty with language or writing please seek help early. See below – Getting Help.

For more information on Academic Integrity for students including explanations, advice and resource see the [UOW Academic Integrity](#) website. The [UOW Library](#) provides detailed referencing guides and information about how to acknowledge the work of others.

Using Generative Artificial Intelligence (GenAI)

GenAI technology (such as ChatGPT or Microsoft Co-pilot) is reshaping the University experience worldwide. UOW & UOWCA is committed to embracing GenAI as a tool to enhance learning experiences and develop vital work-readiness skills. However, misuse or use of GenAI in assessments where prohibited constitutes academic misconduct (as specified by [Academic Integrity and Student Conduct Policy](#)).

It is important that students check if GenAI is permitted for each assessment task and read the [student guidance](#) available on how to use GenAI ethically and critically, equally recognising its capabilities and limitations. For example:

1. **Generative AI is not a substitute for decision-making:** GenAI should complement, not replace, your critical thinking and decision-making skills.
2. **Output quality depends on prompts:** The quality of GenAI outputs is influenced by prompting. Poorly constructed or unclear prompts may generate outputs that are incorrect.
3. **Fact verification is essential:** GenAI outputs can be fabricated, presenting inaccurate information or contain harmful bias. Verify all GenAI outputs against reliable sources.
4. **Protect data and copyright:** Many GenAI technologies collect information in ways that breach privacy and data protection provisions, particularly where the source material is confidential or subject to copyright. Please check the Terms and Conditions of GenAI technologies and if unsure, contact your Subject Coordinator.
5. **Transparency in use:** Where required, you must acknowledge GenAI use, including providing prompt histories and detailing how GenAI was utilised.
6. **Thoughtful and appropriate application:** Be mindful of when and how to use GenAI tools. Assess its appropriateness for each use, and refrain from use when not suitable.

If you have any questions, please contact your Subject Coordinator.

Late Submission of Assessment Tasks

Students must submit all assessment tasks by the due date or penalties will be applied.

If a student submits an assessment after the due date (i.e.: late submission), the assessment will be awarded an original mark (out of the total available mark), and then a penalty will be applied to that original mark in accordance with the table below:

No. of calendar days late	Penalty
1	5% of the original mark.
2	10% of the original mark.
3	15% of the original mark
4	20% of the original mark
5	25% of the original mark
6	30% of the original mark
7	35% of the original mark
More than 7	A mark of zero may be awarded for the assessment.

For details, refer to [Assessment and Examinations Guidelines for Students](#) on the UOW College website.

Academic Consideration

Academic Consideration is a process intended to minimise the impact of serious circumstances beyond a student's control which prevent a student from completing an assessment task by the due date, or from sitting an exam on the scheduled date.

Details can be found in the UOW College [Academic Consideration Policy](#) which is available on the UOW College website.

Assessment Marking and Feedback

Return of Assessment Tasks

UOW College students can normally expect to have results and feedback on assessment tasks within two weeks and before the next assessment task is due. Students will be advised if this timeframe cannot be met due to exceptional circumstances.

Results and feedback for the final assessment in the subject will be withheld until the release of results for the subject. Release of Results dates are published on the [Key Dates](#) calendar for the relevant academic session.

Marks awarded for any assessment task (including examinations) may be subject to scaling at the end of the session at the UOW College Examiners meeting. These will be completed in accordance with the *Assessment Committee and Declaration of Results Procedure*.

For details on subject final grade descriptors, refer to the [Academic Quality Standards](#) which are available on the UOW College website.

Supplementary and Deferred Assessment

Supplementary assessment may be offered to a student who has failed (or “technical” failed) the subject but whose performance is close to that required to pass the subject.

Whether an offer of supplementary assessment will be made is determined by UOW College. The Subject Coordinator will determine the type of supplementary assessment and the submission (or attendance) date, for example, a supplementary exam, a quiz or written task.

A student who passes the supplementary assessment will be awarded the grade of PS (Pass Supplementary) and a maximum mark of 50 for the subject.

Deferred assessment may be offered to a student who has been granted academic consideration for the final or major assessment in the subject. A deferred assessment may be in the form of an extension to submit an assessment task or a new task equivalent to the task that has been missed e.g. a deferred exam.

Review and Appeal (Grades and Other Academic Decisions)

A student may request an explanation of a mark for an assessment task or a final grade for a subject consistent with the student’s right to appropriate and useful feedback on their performance in an assessment task. If a student remains dissatisfied with their grade, they may request a remark of their assessment task using the form provided on the UOW College Website, in line with the UOW College [Remark Request Procedure](#).

A student may also seek further explanation for other academic decisions such as Academic Consideration, Supplementary Assessment, or the application of an academic policy. If a student is not satisfied with the explanation, or has further concerns, they may have grounds for a formal appeal. For further information refer to the [UOW College Student Grievance and Appeals Policy](#) and related procedures, available on the UOW College website.

General

Getting Help

UOW College is committed to assisting you to achieve your goal of university admission. If you are experiencing difficulties with your studies or feel that you simply would like additional support in a certain area, the link below will take you to a range of services offered by UOW College.

<https://www.uowcollege.edu.au/current-students/academic-personal-support/>

Student Feedback

The UOW College is committed to continual improvement in teaching and learning and takes into consideration student feedback from many sources, including direct feedback to tutors and lecturers and from your responses in Subject Evaluation Surveys. Feedback is also used to inform comprehensive reviews of subjects and courses. For more information on Feedback refer the UOW website <https://www.uowcollege.edu.au/current-students/student-feedback/>

Quality Assurance

The UOW College may retain copies of student work to facilitate quality assurance of assessment processes, in support of the continuous improvement of assessment design, assessment marking and for the review of the subject. The UOW College retains records of students' academic work in accordance with the UOW College [Privacy Policy](#), the *State Records Act 1988* and the *Privacy and Personal Information Protection Act 1998*.

Recording of Teaching and Learning Activities

The UOW College supports the recording of lectures and where appropriate, other teaching and learning experiences, as a supplemental study tool to provide students with equity of access, and as a technology-enriched learning strategy to enhance the student experience.

In accordance with the UOW College [Privacy Policy](#) when undertaking our normal teaching and learning activities, the UOW College may collect your personal information. This collection may occur incidentally during the recording of lectures or other learning experiences in recording-equipped venues or online delivery (i.e. when your identity can be ascertained by your image, name, voice or opinion), therefore the UOW College further advises students that:

Lecture and/or online recordings are made available to students, staff, and affiliates, securely on the university's Echo360 ALP (Active Learning Platform) and via the subject Moodle eLearning site.

Recordings are made available only for the purpose for which they were recorded, for example, as a supplemental study tool or to support equity and access to educational resources; and

Recordings are stored securely for up to four years.

If you have any concerns about the use or accuracy of your personal information collected in a recording, you may approach your Subject Coordinator to discuss your particular circumstances. The UOW College is committed to ensuring your privacy is protected. If you have a concern about how your personal information is being used or managed, please refer to the please refer to the UOW College [Privacy Policy](#).

If you make your own recording of a lecture or other learning experience you can only do so with the explicit permission of the subject coordinator, lecturer and those people who are also being recorded. You may only use recordings, whether they are your own or recorded by the UOW College, for your own educational purposes. Recordings cannot be altered, shared or published on another platform, without permission of the UOW College, and to do so may contravene the UOW College [Privacy Policy](#), [Intellectual Property Policy](#), and/or [Academic Integrity and Student Conduct Policy](#). Unauthorised sharing of recordings may also involve a breach of law under the *Copyright Act 1969*.

Learning Analytics

Where Learning Analytics data (such as student engagement with Moodle, access to recorded lectures, University Library usage, task marks, and use of SOLS) is available to the Subject Coordinator, this may be used to assist in analysing student engagement, and to identify and recommend services and support for students who may be in need of assistance. If you have questions about the data the UOW College uses, e.g. how we collect and store data, and how we protect your privacy in the use of this data, please refer to the UOW College [Privacy Policy](#).

Extraordinary Changes to the Subject Outline

In extraordinary circumstances the provisions stipulated in this Subject Outline may require amendment after the Subject Outline has been published. All students enrolled in the subject must be notified and have the opportunity to provide feedback in relation to the amendment, where practicable, prior to the amendment being finalised.

College Policy and Procedures

UOW College policies and procedures can be found at:

<https://www.uowcollege.edu.au/about/governance/policies-procedures/>